

The Spectral Conditional Rate–Distortion Function and the Aging Context

Paper VI skeleton — v0.1, for venue work

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Abstract

Gray’s conditional rate–distortion function — side information at both encoder and decoder — has been known in scalar and finite-vector Gaussian forms since 1972, yet its spectral form for jointly stationary Gaussian pair processes has, to the best of a four-sweep, three-library-check search, never been written down. We supply it: at the work endpoint, $L(D)$ is the equal-slope allocation of per-frequency quadratic-root values over the conditional spectrum, recovering classical Kolmogorov–Pinsker reverse water-filling as the uninformative-context limit; the weighted rate–work version carries per-frequency two-water-level profiles under a single distortion price. En route we prove that the staleness of the conditioning context is an *access-width* phenomenon, not a delay phenomenon — pure delay is information-free under path access (an exact finite- n recoding identity), while time-local access pays a closed staleness tax whose substitution rule depends on the encoder’s epoch of access — and that every observation access class enters through one conditional variance, making the ideal reset work of a stored record against an aging reference an exact Kalman-family curve $W(\Delta)$. [TODO: compress to 150 words for the venue.]

1 Introduction and related work

[TODO: Narrative: Gray 1972/73 → Leiner–Gray 1974 → the static vector line (Gkagkos/Stylianou–Charalambous) → the gap. Attribution spine drafted in go12-process-region-NOVELTY.md; use the demanded sentences verbatim. Position the third-party conditioner (eraser) against Simeone–Permuter, Heegard–Berger, and the vending-machine line.]

2 Model and coordinates

[TODO: Transplant from go12-process-region.tex §The problem; pin: circulant embedding (load-bearing), orthonormal DFT, mean-distortion normalization, Gaussian-record provenance.]

3 Staleness is access width

[TODO: Facts 1–2 (recoding identity; slice dichotomy with the encoder-access substitution rule). Include the R-IND-5 refutation of the first draft’s pairing as a worked

caution (referees like this).]

4 The conditional-variance reduction and the dynamic Landauer curve

[TODO: Theorem 1 ($q_{\mathcal{G}}$); slice/prefix/path = single-sample / Kalman fixed-lag / non-causal Wiener; exact gap $C^{2\Delta}(P_f - q_{\text{path}})$; $W(\Delta)$. Physics scoping remark (del Rio, Sagawa–Ueda, Still et al., delayed-feedback demons).]

5 The spectral conditional RDF

[TODO: Theorem 2-spectral + the weighted theorem (Conjecture 1 closed; the (w, μ) rescope of the “pair of prices”). Anchors as subsections. Convexity proof (verifier’s inverted-quadratic argument) in-line; saturation KKT constant e_k .]

6 The Toeplitz transfer

[TODO: DECISION POINT: either prove Lemmas T1–T4 (joint diagonalization; value continuity; two-sided sandwich; window pinning) — bounded project, $O(1/n)$ numerics clean, n -gap constant to three digits — or state as conjecture with the numerical support and ship circulant-scoped. Draft both texts; Syed picks per venue.]

7 Open problems

[TODO: Process-rate causal object (Conjecture 2', with the Lev–Khina definitional fork made explicit); aging models that break Δ -invariance; AoI–Landauer bridge (metadata-only caveat).]

Reproducibility

Every theorem: sealed preregistration, governed harness, CI re-run on every push, fresh-context adversarial verification. Map: PAPER-VI-VII-EVIDENCE-INDEX.md; archived releases go12-go13-1.0/1.1/1.2 under the concept DOI.

[TODO: Bibliography: transplant the 20-entry list from go12-process-region.tex and extend per the attribution spine.]